Modeling and Simulation of Xe-100 type Pebble-Bed Gas-Cooled Reactor with SCALE

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ABSTRACT

X-Energy's reactor concept, the Xe-100, was selected to be one of two concepts to receive funding to demonstrate its design under the United States (US) Department of Energy's Advanced Reactor Demonstration Program. In order to evaluate the resource requirements, reactor physics calculations are needed to determine the discharge burnup and isotopic content, and fuel residence time. For this purpose, an Xe-100 type pebble-bed high temperature gas reactor model was created in SCALE. The core-averaged and zone-wise equilibrium compositions for the slice and 3D models were obtained following an iterative depletion method developed by the SCALE team for pebble-bed reactor analysis. The SCALE/TRITON transport and depletion calculation for the Xe-100 type slice model indicates that the isotopic concentrations are in good agreement with the results presented in the literature for this reactor type at 1,300 effective full-power days for ^{235}U, ^{236}U, and ^{239}Pu. Sensitivity studies on the pebble location in the core were performed using new capabilities in SCALE/ORIGAMI for pebble-bed depletion intended for inclusion in SCALE Version 7. These results show that the pebbles that travel closer to the reflector for the entire burn history experience a higher burnup due to the higher thermal to fast flux ratio than pebbles that travel through the middle of the core due to the lower thermal to fast flux ratio.

KEYWORDS: SMR, advanced reactors, pebble-bed, depletion, SCALE, ORIGAMI

1. INTRODUCTION

The United States (US) Department of Energy (DOE) started an Advanced Reactor Demonstration Program (ARDP) to accelerate the commercialization of advanced reactor concepts. In 2020, DOE identified two advanced reactor concepts, one of which is X-Energy's Xe-100 to receive funding for demonstrating their advanced reactor concept. In a complementary effort, the Systems Analysis and Integration (SA&I) Campaign within the DOE Office of Nuclear Energy has been performing transition and deployment scenario studies to evaluate the transition of the current commercial fleet of light water reactors (LWRs) to a future fleet of advanced reactors consisting of a mix of ARDP-type reactor concepts and advanced LWRs [1]. To accurately evaluate the front- and back-end resource requirements, reactor physics calculations were performed to determine parameters such as the discharge burnup, discharge isotopic content, and fuel residence time.

The isotopic inventories of spent nuclear fuel have a significant impact on the analysis of the overall fuel cycle. Spent fuel isotopic compositions vary based on the conditions the fuel experienced during its lifetime, including the total burnup time and neutron flux. Within this project, our goals were to construct an Xe-100 type reactor model based on data from publicly-available literature, estimate a core-averaged equilibrium fuel composition, and from this calculate the relative sensitivity of discharged isotopic compositions for major actinide species to cumulative burnup and location within the core. To this end, SCALE/TRITON was used to create the reactor models. Initial determination of the equilibrium composition was performed with SCALE/ORIGEN and an iterative scheme that was developed by Bostelmann et. al. for pebble-bed analyses detailed in [2,3]. Sensitivity studies were conducted using newly-developed SCALE/ORIGAMI features designed to address pebble-bed depletion [3,4] which is discussed further in Section 2.

2. METHODS

Using methods previously developed for modeling pebble-based systems [2,4], SCALE/ORIGEN reactor data libraries libraries were generated at different temperatures for pebbles at varying radial locations in the core using SCALE/TRITON. These libraries were then used to perform a sensitivity study on the relative impact of the pebble position on the fuel composition during core transits using a newly-refactored version of ORIGAMI introduced in a pre-release version of SCALE7. The new ORIGAMI includes features designed to perform rapid depletion analysis for flowing pebble-bed systems [3,4]. In addition to the SCALE/ORIGEN library generation, SCALE/TRITON was used to obtain the data needed for the iterative scheme, which eventually resulted in the generation of equilibrium fuel compositions for a full core model.

2.1. ORIGAMI Overview

Introduced in SCALE Version 7.0-beta1, the latest version of SCALE/ORIGAMI is designed to analyze both static fuel assemblies and flowing pebble-bed systems [3,4]. It provides a simple, fast, and lightweight way to determine zone-wise isotopic inventories, discharge pebble characteristics, and perform sensitivity and uncertainty analysis.

This new version of ORIGAMI models core depletion via a series axial zones, called “transit zones” in flowing-fuel systems. Each transit zone can contain multiple radial nodes, each with unique parameters. Multiple transit zones are stacked together to represent the path of the pebble through the core. After completing one pass through the core, the masses in the radial zones are summed, the depleting material is decayed for a user-defined downtime, and then the material masses are redistributed according to the radial mass distributions in the next transit zone. Multiple passes through the core can be simulated to model the complete lifetime of fuel pebbles in a given core. Figure 1 further illustrates this analysis workflow with the new ORIGAMI implementation used for this work.

2.2. Full-Core and Slice Model Descriptions

The model developed for this study was based on the Xe-100, a 165 MWth pebble-bed HTGR as described in [5]. When it was not possible to determine the dimensions of the Xe-100 from [5], an assumption was made using information for the PBMR-400 [6].

Two types of models were created in SCALE: (1) a 3D full-core model (Figure 2a), used to determine axial power profiles and equilibrium core compositions, and 2) a 2D slice model

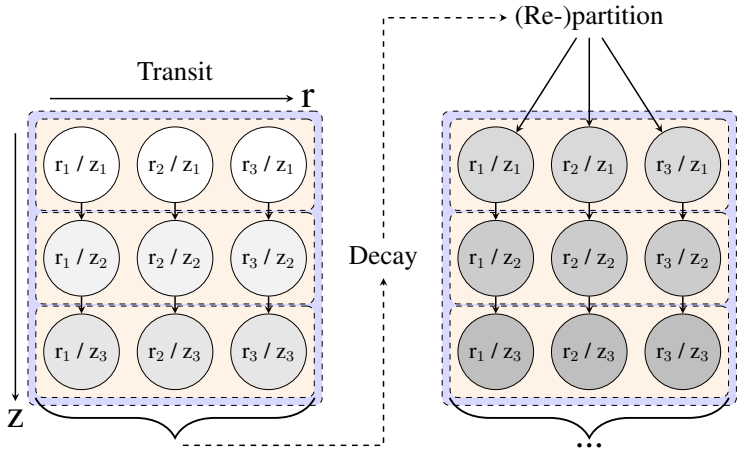
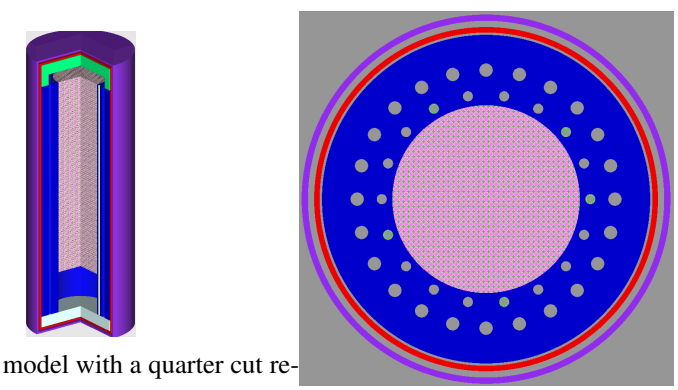


Figure 1: Flowchart of the SCALE/ORIGAMI depletion method for flowing-pebble systems, from [3]. Pebbles (circles) are represented as radial channels within a "transit zone" (light orange), which are grouped together to represent a pass (transit) through the core (light blue). Upon each transit through the core, masses for each radial zone to be retained are summed and then re-distributed across the radial regions for the next core transit.

(Figure 2b), a 10 cm thick cross section of the active fuel region of the full-core model used to determine radial power profiles and SCALE/ORIGEN cross section libraries. The radial power profile (shape) was obtained from the slice model, which was subdivided into 3 regions, to only provide the approximate radial power "shape."



(a) Full-core model with a quarter cut removed

(b) Top-down view

Figure 2: 3-D full core model (with one axial zone) and 2D axial slice model used for SCALE/ORIGEN library generation. Helium is shown at reduced opacity in the 3-D models for clarity.

These models use the following assumptions:

- Material temperatures and specific power are constant throughout the model
  - A notable exception to this is in the ORIGAMI input, where axial and radial zones vary in fuel and moderator temperatures, and power.

- The exact number of helium riser channels in the reflector is not explicitly given in [5]. To determine the number of risers for these models, it is assumed that the center-to-center distance between riser channels (measured as the arc length) is the same as the spacing found in the PBMR-400 [6].

- Five of the nine main control rods are fully inserted in an alternating pattern, while the other 4 control rods and the 9 emergency rods are fully withdrawn, and modeled as an empty channel filled with helium.

- Pebbles make six passes through the core in their lifetime.

2.3. Equilibrium Core Composition

The equilibrium core composition for the Xe-100 is not publicly available. An iterative method was used to determine the equilibrium full core composition, using the ”slice” model described in full in [2], it will also be summarized here.

1. An initial core of all fresh pebbles is assumed in the “slice” model.

2. Five “surrogate” pebbles are selected for depletion, spaced radially throughout the slice to capture the radial variation in the neutron flux shape.

3. These pebbles are depleted using SCALE/TRITON at a constant power of $105.7 \frac{\mathrm{MW}}{\mathrm{MTU}}$ for 1,561 days to achieve a discharge burnup of 165 $\frac{\mathrm{GWd}}{\mathrm{MTIHM}}$.

4. SCALE is used to interpolate the 1-group cross section libraries for depletion substeps at the beginning/end of user-specified axial zones and passes.

5. SCALE/ORIGEN then determines the zone- and pass-wise isotopic compositions in each axial zone. For example, if there are three axial zones, and the pebbles make two passes, there will be six distinct isotopic composition (one for each pass for in each axial zone).

6. The pass-wise isotopic compositions are averaged to determine the zone-wise compositions for the reactor, and these zone-wise compositions are averaged to determine the core-average isotopic composition.

7. This new core-averaged composition replaces the fresh fuel composition in all the pebbles except the five surrogate pebbles in the SCALE/TRITON “slice” model. The surrogate pebbles still have fresh fuel compositions to allow for depletion from 0 to 165 $\frac{\mathrm{GWd}}{\mathrm{MTIHM}}$ amongst pebbles surrounded by the new core-averaged equilibrium composition.

8. This process repeats (steps 3 through 8) until the isotopes of interest (i.e. ^{235}U, ^{239}Pu) converge.

This analysis uses six axial zones of equal volume, and assumes that each pebble makes six passes over its lifetime.

2.4. Temperature-Dependent Cross Section Library Generation For SCALE/ORIGAMI Calculations

Before performing SCALE/ORIGAMI sensitivity analyses, it is necessary to create the temperature-dependent HDF5 libraries for the simulations. To do this, the slice model is used once again, however, the slice is split into three radial zones to create cross sections that would capture differences in the cross sections due to differences in the radial flux. These three radial zones are defined such that they have equal volumes. Fresh surrogate pebbles, as in the iterative equilibrium composition generation method, are placed in each zone. The non-depleting pebbles are composed of the core-averaged equilibrium composition obtained using the methodology outlined in the previous subsection. Both depleting and non-depleting pebbles are uniquely defined in the radial zones, resulting in six separate sets of materials in the slice model (3 depleting and 3 non-depleting materials). SCALE/TRITON inputs for the "slice" model with material compositions for fuel and moderator temperatures of 600, 750, and 900 K were created to make a series of slice model variations. Upon completion of each SCALE/TRITON calculation, the one-group cross section files (`*.f33`) are tagged and compiled into an HDF5 library.

2.5. Determining Radial and Axial Power Profiles for SCALE/ORIGAMI Calculations

The radial and axial power profiles in the Xe-100 are required to provide the shape of the power in SCALE/ORIGAMI. The radial power profile can be determined using the output files generated during the SCALE/TRITON simulations to make the HDF5 cross section library. These output files were used to determine the radial power profile in the three separate radial zones.

The axial power profile is determined by creating and running a SCALE/TRITON full-core model with six axial fuel regions of equal volumes. The material compositions for each of these six zones are extracted from the zone-wise compositions. The resulting zone-wise powers in the SCALE/TRITON output is used to determine the axial power shape for SCALE/ORIGAMI. The ratio of the power profile from the nominal power in each of the six zones are shown below with Zone 1 being the first axial zone at the top and Zone 6 being the last axial zone at the bottom of the core:

Table I: Zone-wise axial power profile and reflector temperatures

Zone	Normalized power profile	Reflector temperature [K]
1	1.16	623.15
2	1.74	623.15
3	1.50	623.15
4	0.949	623.15
5	0.469	648.15
6	0.158	723.15

The power profile shows that the power peaks towards the top of the core since the pebbles flow from the top of the reactor to the bottom. For this full core model with equilibrium compositions, the k_{eff} is 0.95821 +/- 0.00014. The eigenvalue (k_{eff}) is less than one because the goal of this work

was not to optimize for this global parameter by adjusting the control rods or material impurities in the reflector or other regions of the core. However, optimizing for the eigenvalue could be included in future work if it is of value and interest. For the purposes of this work, the depletion calculations are performed and only require the specific power which is used to normalize the flux.

Table II: Radial power profile & temperatures by axial zone

Zone	**Normalized radial power profile**			**Fuel temperature (K)**		
	Outer	Middle	Inner	Outer	Middle	Inner
1	0.21141[1]	0.33199[1]	0.44535[1]	623.15	823.15	873.15
2				773.15	823.15	873.15
3				900.0[2]	900.0[2]	900.0[2]
4				900.0[2]	900.0[2]	900.0[2]
5				900.0[2]	900.0[2]	900.0[2]
6				900.0[2]	900.0[2]	900.0[2]

[1] The same radial power distribution is used for all axial nodes

[2] Separate analysis indicated a higher fuel temperature for these regions, however the maximum fuel temperature permutation used for library generation in this study was 900 K. Thus, the upper limiting value was used for the fuel temperature in these zones.

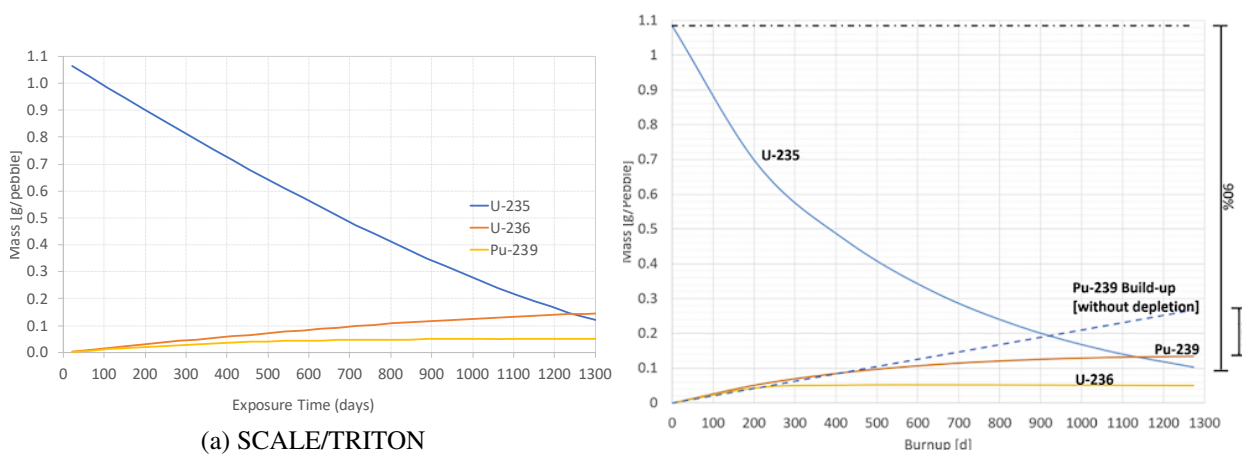
3. RESULTS AND DISCUSSION

It was important to determine whether the SCALE/TRITON and SCALE/ORIGAMI were able to determine the discharge isotopic vectors consistently with publicly released information in [5] and [7] thereby increasing confidence in the SCALE models and results. Figure 3 shows select uranium and plutonium isotopes as a function of exposure time generated by SCALE/TRITON next to the results published in [5]. The plots show that the discharge ^{235}U content (at 1,300 days of total burn time) from SCALE/TRITON is about 0.12 $\frac{\text{grams}}{\text{pebble}}$, whereas it is about 0.1 $\frac{\text{grams}}{\text{pebble}}$in Reference [5]. The ^{235}U depletion rate in SCALE has a linear profile while the paper published by X-Energy [5] does not have this profile. For example, within the SCALE models, a constant pebble power is used throughout each core transit when in reality the pebble specific power will be higher than the average the beginning of life and below the average power as it approaches its discharge burnup. When comparing the ^{236}U and ^{239}Pu isotopes, the results generated by SCALE/TRITON and in [5] are reversed. However, SCALE/TRITON results for ^{239}Pu and ^{236}U curves match the results shown in a presentation to the MeV school at Argonne National Laboratory [7], and leads the authors to believe that there is a minor transcription issue in Reference [5]. Since, the SCALE/TRITON results agree with [7], it gives further confidence in the pebble depletion methodology set up in SCALE/TRITON.

After confidence in the SCALE/TRITON model was achieved with the comparison to publicly available results, the discharge isotopic concentrations of these same isotopes were analyzed using SCALE/ORIGAMI, in a series of four runs, labeled “control,” “inner,” “middle,” and “outer.” The “control” case models the Xe-100 type reactor with three radial zones: the inner (center-most), middle, and outer (nearest reflector) zones. It assumes that there is an equal probability that the pebble will flow through one of the three radial regions. The “inner,” “middle,” and “outer” cases considered a pebble exclusively confined to one of these three regions in each transit through the core. The discharge isotopics for the “inner,” “middle,” and “outer”

cases change as a result of the spectral change the pebble undergoes exclusively in each of these regions. The results for the "Control" case (see Table III) indicates that the isotopic inventory is similar to what was generated with SCALE/TRITON. The ^{235}U, ^{236}U and ^{239}Pu at discharge at 1,590 days in SCALE/ORIGAMI is approximately 0.05, 0.15, and 0.049 $\frac{\text{grams}}{\text{pebble}}$ whereas with TRITON at 1,561 days, the corresponding results were 0.04, 0.15, and 0.048 $\frac{\text{grams}}{\text{pebble}}$.

Skutnik previously identified observed differences in the PBMR-400 pebble discharge vectors driven by shifts in the neutron spectrum relative to the pebble's proximity to reflector regions [4]. As seen in Figure 4, the outer (blue) radial zone has a higher ratio of thermal to fast neutrons than the inner and middle zones due to its proximity to the reflector. Thus, the "Outer" zone near the reflector sees a relatively higher ratio of thermal to fast neutrons compared to the "Inner" zone, therefore achieving a higher fission rate (and thus higher burnup). This is evident when viewing Table VI for the pebble that travels exclusively through the "outer" region of the core. The "Inner" and "Middle" zones have similar discharge ^{235}U content of 0.048 and 0.046 $\frac{\text{grams}}{\text{pebble}}$, respectively (see Tables IV and V). However, for the "Outer" zone which experiences a higher thermal to fast ratio (near the reflector), the ^{235}U content is 0.0296 $\frac{\text{grams}}{\text{pebble}}$. Therefore, the longer a pebble stays near the reflector, the higher the burnup, whereas the closer it stays to the center of the core, the lower the burnup. These sensitivity studies using ORIGAMI are valuable and more of these analyses can be performed in the future for these types of reactors in SCALE since the workflow has been successfully demonstrated for pebble-bed reactors.



(a) SCALE/TRITON

(b) X-Energy [5]

Figure 3: Select isotopic compositions as a function of exposure time (days), as calculated using SCALE/TRITON and those reported by X-Energy [5]. Results are reported in mass per 7 gHM pebble.

4. CONCLUSIONS

Full-core and slice models of an Xe-100 type pebble bed reactor were created in SCALE. SCALE/TRITON and SCALE/ORIGEN were used to generate the equilibrium compositions for these models. The SCALE/TRITON results at 1300 days (exposure time) are consistent with the results presented by Mulder [7]. The ^{235}U isotopic composition from SCALE/TRITON has a linear trendline over the exposure time, which alludes to differences in the models and methods between SCALE/TRITON and those used by X-Energy to generate their results [5,7].

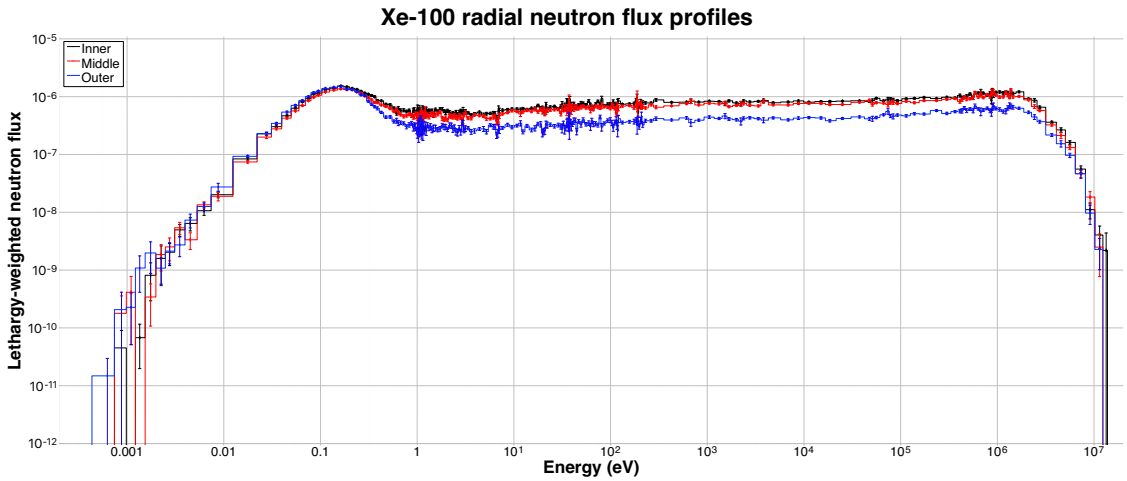


Figure 4: 252-group neutron energy spectrum for the three radial zones modeled.

Table III: Isotopic inventories over time $\frac{\text{grams}}{\text{pebble}}$: Control case

Exposure Time (Days)	^{235}U	^{236}U	^{238}U	^{238}Pu	^{239}Pu	^{240}Pu	^{241}Pu
0	1.085	0	5.915	0	0	0	0
260	0.849	0.0388	5.877	1.692×10^{-5}	2.619×10^{-2}	3.12×10^{-3}	6.689×10^{-4}
525	0.630	0.0737	5.832	1.426×10^{-4}	4.090×10^{-2}	9.352×10^{-3}	3.397×10^{-3}
790	0.432	0.104	5.776	5.444×10^{-4}	4.747×10^{-2}	1.579×10^{-2}	7.736×10^{-3}
1060	0.261	0.128	5.704	1.501×10^{-3}	4.943×10^{-2}	2.080×10^{-2}	1.215×10^{-2}
1320	0.129	0.143	5.605	3.368×10^{-3}	4.921×10^{-2}	2.382×10^{-2}	1.510×10^{-2}
1590	0.0475	0.147	5.468	6.240×10^{-3}	4.854×10^{-2}	2.492×10^{-2}	1.634×10^{-2}

Table IV: Isotopic inventories over time $\frac{\text{grams}}{\text{pebble}}$: Inner case

Exposure Time (Days)	^{235}U	^{236}U	^{238}U	^{238}Pu	^{239}Pu	^{240}Pu	^{241}Pu
0	1.085	0	5.915	0	0	0	0
260	0.848	0.0392	5.876	1.346×10^{-5}	2.805×10^{-2}	3.243×10^{-3}	5.900×10^{-4}
525	0.630	0.0743	5.829	1.306×10^{-4}	4.296×10^{-2}	9.825×10^{-3}	3.512×10^{-3}
790	0.433	0.105	5.772	5.274×10^{-4}	4.952×10^{-2}	1.649×10^{-2}	8.308×10^{-3}
1060	0.263	0.129	5.699	1.488×10^{-3}	5.085×10^{-2}	2.139×10^{-2}	1.3146×10^{-2}
1320	0.130	0.144	5.599	3.387×10^{-3}	5.057×10^{-2}	2.397×10^{-2}	1.619×10^{-2}
1590	0.0483	0.148	5.463	6.396×10^{-3}	5.008×10^{-2}	2.523×10^{-2}	1.713×10^{-2}

Once the equilibrium core compositions were generated, they were used in the slice models for varying fuel and moderator temperatures to generate a set of temperature-dependent one-group

Table V: Isotopic inventories over time in $\frac{\text{grams}}{\text{pebble}}$: Middle case

Exposure Time (Days)	^{235}**U**	^{236}**U**	^{238}**U**	^{238}**Pu**	^{239}**Pu**	^{240}**Pu**	^{241}**Pu**
0	1.085	0	5.915	0	0	0	0
260	0.848	0.0392	5.876	1.356×10^{-5}	2.778×10^{-2}	3.094×10^{-3}	5.688×10^{-4}
525	0.629	0.0743	5.830	1.291×10^{-4}	4.310×10^{-2}	9.456×10^{-3}	3.372×10^{-3}
790	0.431	0.105	5.774	5.165×10^{-4}	4.979×10^{-2}	1.611×10^{-2}	7.982×10^{-3}
1060	0.260	0.129	5.701	1.468×10^{-3}	5.172×10^{-2}	2.125×10^{-2}	1.274×10^{-2}
1320	0.127	0.145	5.602	3.379×10^{-3}	5.144×10^{-2}	2.418×10^{-2}	1.595×10^{-2}
1590	0.0463	0.148	5.465	6.380×10^{-3}	5.092×10^{-2}	2.507×10^{-2}	1.718×10^{-2}

Table VI: Isotopic inventories over time in $\frac{\text{grams}}{\text{pebble}}$: Outer case

Exposure Time (Days)	^{235}**U**	^{236}**U**	^{238}**U**	^{238}**Pu**	^{239}**Pu**	^{240}**Pu**	^{241}**Pu**
0	1.085	0	5.915	0	0	0	0
260	0.847	0.0384	5.883	9.981×10^{-6}	2.314×10^{-2}	2.478×10^{-3}	3.714×10^{-4}
525	0.624	0.0736	5.845	9.680×10^{-5}	3.665×10^{-2}	8.074×10^{-3}	2.363×10^{-3}
790	0.420	0.105	5.796	4.025×10^{-4}	4.268×10^{-2}	1.438×10^{-2}	6.033×10^{-3}
1060	0.241	0.130	5.730	1.193×10^{-3}	4.481×10^{-2}	1.974×10^{-2}	1.013×10^{-2}
1320	0.105	0.145	5.635	2.889×10^{-3}	4.462×10^{-2}	2.295×10^{-2}	1.316×10^{-2}
1590	0.0296	0.148	5.495	5.650×10^{-3}	4.432×10^{-2}	2.374×10^{-2}	1.431×10^{-2}

cross sections. These cross sections were compiled together to generate temperature-dependent cross section libraries that can be used to perform quick sensitivity calculations with a new flowing-pebble depletion capability for ORIGAMI (to be included in the SCALE 7.0 release).

ORIGAMI inputs were run for four cases: a “control” model in which pebbles are distributed across three radial zones during each pass through the core with equal probability, and then three separate runs (“inner”, “mid”, and “outer”) in which pebbles travel exclusively through one of the three radial zones throughout each pass through the core. The purpose of this analysis was to provide for a bounding sensitivity analysis as to the effects of the pebble location history within the core, as the neutron spectrum shape has been previously shown to be highly sensitive to the pebble’s radial proximity to the graphite reflector.

The results generated in this paper from SCALE Xe-100 type pebble bed models indicate that the methods and models presented in this work are consistent with those reported in [5] and [7]. These models can therefore be used to perform other scoping studies with SCALE. Future work could include expanding the cross section library to include a wider range of temperature-dependent cross sections since the reference library used only extended to a reflector temperature of 900 K. Using the steps detailed in Section 2, the temperature range can be expanded to include data points at 1050 and 1200 K, which would cover the entire operating temperature range in a pebble-bed reactor. Additionally, while the four ORIGAMI runs described

here cover the most basic cases, there is still room to analyze a series of cases that perturb the axial and radial power profiles, fuel and reflector temperatures, and the pebble radial distributions in order to produce a full sensitivity profile of the discharged isotopic inventories. These sensitivity studies would be more easily performed since the models and inputs have now established in SCALE using publicly available information.

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REFERENCES

[1] T. K. Kim, S. Richards, A. Cuadra, R. Hays, J. W. Bae, and E. Davidson. "Report on Fuel Cycle Facility Requirements for Deployment of Demonstration Reactors and Potential Evolutionary Fuel Cycle Scenarios." Technical Report ANL/NSE-22/39, Argonne National Laboratory (2022).

[2] F. Bostelmann, C. Celik, R. F. Kile, and W. A. Wieselquist. "SCALE Analysis of a Fluoride Salt-Cooled High-Temperature Reactor in Support of Severe Accident Analysis." Technical Report ORNL/TM-2021/2273, Oak Ridge National Laboratory (2022).

[3] S. Skutnik, F. Bostelmann, and W. Wieselquist. "Rapid Depletion Analysis of Flowing-Pebble Reactor Systems at Equilibrium using SCALE." In *International Conference on Physics of Reactors 2022 (PHYSOR 2022)* (2022).

[4] S. E. Skutnik and W. A. Wieselquist. "Assessment of ORIGEN Reactor Library Development for Pebble-Bed Reactors Based on the PBMR-400 Benchmark." Technical Report ORNL/TM-2020/1886, Oak Ridge National Laboratory, Oak Ridge, TN (2021).

[5] E. Mulder and W. Boyes. "Neutronics characteristics of a 165 MWth Xe-100 reactor." *Nuclear Engineering and Design*, **volume 357**, p. 110415 (2020). URL https://www.sciencedirect.com/science/article/pii/S0029549319304467.

[6] International Atomic Energy Agency (IAEA). *Evaluation of High Temperature Gas Cooled Reactor Performance: Benchmark Analysis Related to the PBMR-400, PBMM, GT-MHR, HTR-10 and the ASTRA Critical Facility*. IAEA-TECDOC-1694. IAEA, Vienna, Austria (2013).

[7] E. Mulder. "Xe 100: Aspects of Multi Scale, Multi Physics Modeling." (2021). Presentation to MeV School.